MandiQ: AI Mandi Price Intelligence & Smart Agriculture Platform

Detailed Project Explanation in Hinglish | Architecture, Frontend, Backend, ML Models, APIs & Live Deployments

Project: MandiQ (Price Forecasting & Smart Sell)	**Target Audience:** Small Farmers & Mandi Traders
Live Frontend: https://mandi-q.vercel.app	**Live Backend:** https://mandiq-3l0e.onrender.com
GitHub Repository: https://github.com/Sayalipatil-17/MandiQ	**Tech Stack:** React + Vite, FastAPI, MongoDB, XGBoost/LightGBM

1. MandiQ Overview & Problem Statement (MandiQ Kya Hai aur Kyun Hai?)

Bharat ke kisaano ke saamne sabse badi problem hoti hai **Price Volatility (daam me achanak utar-chadaav)** aur **Market Information Asymmetry**. Kisaan jab apni fasal lekar mandi pahunchta hai, tab use pata chalta hai ki aaj daam gir chuke hain, jisse use majbooran aadhaye daamo me bechna padta hai. Saath hi aadhate/middlemen unse bhaari commission kaat lete hain, aur kisaano ko ye bhi nahi pata hota ki paas ki doosri mandi me wahi fasal mehengi bik rahi hai.

MandiQ is problem ko 4 tareeqon se solve karta hai:

- **Price Prediction Engine:** Agmarknet ke historical data aur machine learning ki madad se agle 7 se 30 dino ka mandi bhav pehle hi bata deta hai.
- **Sell / Hold AI Recommendation:** Kisaan ko clear advice milti hai ki fasal aaj bechni chahiye ya hold karna faydemand rahega.
- **Cross-Mandi Arbitrage (Transport Deduction):** Alag-alag mandiyo (jaise Azadpur vs Keshopur vs Prayagraj) ke daam compare karke, transport kharcha minus karke batata hai ki kahan sabse zyada net profit hoga.
- **SmartSell Direct Buyer Demand & Calculator:** Mandi ke 10-12% traditional deductions (commission, palledari, katai) ke muqable direct buyers (agro-exporters, retail chains) se connect karke kisaan ka munafa badhata hai.
- **Vernacular Language Support:** 4 bhaashaon (Hindi, English, Punjabi, Marathi) me voice/text NLP support deta hai.

2. End-to-End System Architecture (Pura System Kaise Kaam Karta Hai)

MandiQ ek modern decoupled client-server architecture follow karta hai:

Component	Technology	Function / Role
Frontend Client	React 18 + Vite + TS + Tailwind + Capacitor	Responsive Web & Android app UI, charting, multilingual interface, local caching.
Backend API	FastAPI (Python 3.10+)	High performance async REST APIs, rate limiting, background tasks, auth & routing.
Database Layer	MongoDB Atlas (Cloud) + SQLite3 (Fallback)	Stores price records, users, crop preferences, alerts, OTP sessions, buyer demands.
ML / Prediction	Mean-Reversion + 4-Model Ensemble	Random Forest, Extra Trees, XGBoost, LightGBM + Reversion engine with 69 features.
Data Pipeline	Agmarknet Scraper + 2Captcha	Automated daily scrape of government mandis, automated captcha resolution.
Push Notifications	OneSignal REST API + Fast2SMS	Real-time browser push notifications and SMS alerts for price target triggers.

3. Frontend Architecture & Screen Walkthrough

Frontend ko React + Vite ke sath banaya gaya hai. Codebase me **React Router v7**, **Lucide Icons**, aur **Recharts** ka use kiya gaya hai. App me total 12+ screens hain jo farmers ke workflow ko step-by-step complete karti hain:

Screen Name	Key Functions & Details
1. SplashScreen	App khulte hi check karta hai ki user ke paas valid JWT session hai ya nahi. Agar session hai to direct HomeScreen, warna LoginScreen par redirect karta hai.
2. LoginScreen	Farmer apna 10-digit mobile number daalta hai. Validation ke baad backend ko /api/auth/send-otp request jati hai.
3. OtpScreen	6-digit SMS OTP verify karta hai. Testing aur offline evaluation ke liye developer bypass code (DEV_OTP_BYPASS) bhi support karta hai.
4. Onboarding Wizard	Naye users ke liye 4-step wizard: **UserTypeScreen** (Farmer/Trader), **NameScreen**, **LocationScreen** (State, District, Village), aur **CropSelectionScreen** (Tomato, Potato, Onion, Spinach).
5. HomeScreen	Core dashboard: Aaj ka modal price, 7-day price forecast strip (Actual aur Predicted ka seamless hybrid), interactive price graph, best nearby mandi compare card, aur quick actions.
6. PredictionScreen	Detailed 7-day aur 30-day forecast chart with 80% confidence interval (upper & lower price bands). AI Recommendation card (Sell Now vs Hold for X days) aur user prediction accuracy feedback system.
7. SmartSellScreen (New)	SmartSell Calculator: Direct buyers (Agro Exports, Wholesalers) ki demand se match karta hai. Quality grade (A: 1.00x, B: 0.92x, C: 0.82x), Mandi deductions vs Direct selling profit breakdown aur historical sale record logging.
8. MandiCompareScreen	Alag-alag mandis (Azadpur, Keshopur, Shahdara, Prayagraj) ke live prices fetch karke, distance ke mutabiq transport cost minus karta hai aur batata hai ki 'Net Profit' kahan sabse zyada milega.
9. AlertsScreen	Custom price alerts banata hai (jaise 'Tomato > Rs 2500 hone par alert karo' ya 'Best Day to Sell alert'). OneSignal web push aur SMS notification triggers.
10. PastTrendScreen	Historical multi-year trends, monthly seasonal averages (kaunse mahine me sabse mehenga bikta hai), aur price volatility analysis.
11. MandiInfoScreen	Mandi coordinates, geographic maps, contact directory, address aur APMC helpline numbers.
12. ProfileScreen	User profile details, dynamic language switcher (Hindi, English, Punjabi, Marathi), notification toggles aur secure logout.
SupportChat Widget	Floating multilingual chatbot assistant. Isme offline keyword dictionary + fuzzy typo matching NLP engine hai jo 4 bhashaon me farmer ke sawalon ka instant jawab deta hai.

4. Backend Microservices, Scraping & Cron Jobs

Backend Python **FastAPI** framework me asynchronous structure par build hai. Iske main modules:

- **Agmarknet Daily Scraper (daily_scrape.py):** Agmarknet portal se live daily arrival quantity aur modal price pull karta hai. Captcha bypass ke liye 2Captcha API aur local image processing use karta hai.
- **APScheduler Automated Jobs:**
  - *Daily 6:00 AM IST:* Sirf naya incremental data pull karke database update karta hai.
  - *Sunday 2:00 AM IST:* Sabhi crops aur mandis ke models ko automated retrain karta hai (run_pipeline.py).
  - *Daily 9:00 AM & 6:00 PM IST:* User alerts evaluate karke OneSignal push bhejta hai.
- **Rate Limiter Middleware (rate_limiter.py):** In-memory sliding window algorithm jo DDoS aur brute-force OTP requests ko block karta hai (max 5 OTP/day per IP).
- **PDF & CSV Parsers (pdf_parser.py, csv_parser.py):** Government APMC reports (PDF ya CSV) upload hone par automated parse karke database me normalize karta hai.

5. Machine Learning & Forecasting Models

Mandi price prediction me simple time-series models fail ho jaate hain kyunki agriculture prices me sudden supply shocks, harvest seasonality aur weather ka bada asar hota hai. MandiQ me do-tier machine learning system use hota hai:

A. Primary Predictor: Mean-Reversion Model (mandiq_reversion.py)

Ye production ka primary model hai. Empirical testing me iska **All-Day MAPE sirf 7.3%** aaya hai. Ye 3-day rolling average (ma3) se price deviation calculate karta hai aur 3 confidence tiers return karta hai:
• **High Confidence (Deviation ≥ 6%):** ~65% historical direction accuracy.
• **Medium Confidence (Deviation 4% - 6%):** ~62% historical direction accuracy.
• **Low Confidence (Deviation 2% - 4%):** ~58% accuracy.
• **Dead Zone (< 2% Deviation):** No clear move — Hold advice generate karta hai.

B. Secondary 4-Model Weighted Ensemble (model_trainer.py)

Deep multi-day recursive forecasting ke liye 4 models ka ensemble chalta hai:
1. **Random Forest Regressor (40% Weight):** Outliers aur sudden noise ko absorb karta hai.
2. **Extra Trees Regressor (40% Weight):** Variance drop karke smooth forecast banata hai.
3. **XGBoost Regressor (10% Weight):** Sharp trend changes aur gradient boosting handle karta hai.
4. **LightGBM Regressor (10% Weight):** Fast tree-leaf splitting aur seasonal interactions capture karta hai.

C. Feature Engineering Pipeline (69 Total Features)

• **Autoregressive Lags:** Past 1, 2, 3, 5, 7, 10, 14, 21, 28, 30-day historical prices.
• **Rolling Statistics:** 7, 14, 30, 60, 90-day rolling mean, std (volatility), min, max, CV.
• **Arrival Dynamics:** Mandi arrival volume (MT), log arrival, price × supply interaction ratio.
• **Calendar & Seasonality:** Day-of-year cyclic sin/cos encoding, month, Indian harvest seasons (Kharif, Rabi, Zaid).
• **Weather Lag Factors:** Temperature min/max, rainfall, humidity with harvest delay offsets.

6. Database Architecture (Hybrid MongoDB & SQLite)

database.py me hybrid storage manager banaya gaya hai. Agar cloud connection string MONGO_URI environment variable me defined hai, to app automatic **MongoDB Atlas** se connect ho jata hai. Agar internet ya cloud key nahi hai, to local file Backend/data/mandiq.db (SQLite3) par automatic fallback ho jata hai.

Collection / Table	Key Fields & Schema Description
price_records	commodity, market, date, modal_price, min_price, max_price, arrivals_in_qtl. Primary historical price repository.
users	user_id, mobile_number, name, role (farmer/trader), state, district, village, preferred_crop, language.
alerts	alert_id, user_id, crop, market, target_price, direction (above/below/best_day), status (active/triggered).
otps	mobile, otp_hash, expires_at (5 mins), attempts. Temporary authentication sessions.
buyer_demands	buyer_id, buyer_name, crop, grade, required_qty_kg, offered_rate_per_kg, location.

7. Live Production Infrastructure & Deployment Setup

Project completely cloud-hosted aur live production me chal raha hai:

• **GitHub Central Repository:** https://github.com/Sayalipatil-17/MandiQ (Branch: main). Dono Frontend aur Backend isi repository se synchronized hain.

• **Backend Service (Render):** Hosted at https://mandiq-3l0e.onrender.com/. Root directory Backend, Python 3.10+, start command: uvicorn main:app --host 0.0.0.0 --port $PORT. MongoDB Atlas se securely linked.

• **Frontend App (Vercel):** Hosted at https://mandi-q.vercel.app/. Auto-build trigger on git push, vercel.json rewrites for Single Page App client routing.

• **Production Environment Configuration:**
  • VITE_API_URL = https://mandiq-3l0e.onrender.com (Vercel Build Setting)
  • MONGO_URI = MongoDB Atlas cloud cluster URI
  • JWT_SECRET_KEY = HMAC-SHA256 signature token
  • FAST2SMS_API_KEY & TWOFACTOR_API_KEY = SMS gateways
  • TWOCAPTCHA_API_KEY = Agmarknet automated scraping solver

8. Summary & Key Impact for Farmers

MandiQ sirf ek price tracking app nahi hai, balki ek complete **decision-support ecosystem** hai jo chote kisaano ko mandi ke daamo par aage ki visibility deta hai. Iske zariye kisaan behtar mandi chun sakte hain, middlemen ke faaltu commission bacha sakte hain, aur sahi samay par fasal bechkar **15% se 25% tak ka extra munafa** kama sakte hain.